WE-DETR: Wavelet-Enhanced Multiscale Feature Compensation Detection Transformer for Remote Sensing Small Objects

Yu Song, Weiguo Zuo, Ronghua Shang, *Senior Member, IEEE*, Haixiao Zhuang, Chen Zheng, *Member, IEEE*, Zhenzhen Liu, and Xiaohui Yang

***Abstract*—Small object detection in remote sensing imagery remains challenging due to the extremely limited pixel footprint of targets and the severe loss of fine-grained spatial details caused by multistage downsampling. To address these issues, we propose WE-DETR, a wavelet-enhanced multiscale feature compensation detection transformer. WE-DETR establishes a remote sensing-oriented progressive high-frequency (HF) feature compensation mechanism, which aims to preserve, enhance, fuse, and propagate discriminative structural cues throughout the detection pipeline. In particular, wavelet decomposition is introduced at the early backbone stage to preserve fine-grained details before downsampling-induced degradation occurs. A low-frequency-guided structural enhancement strategy is further developed to selectively strengthen informative HF components. The enhanced frequency-domain representations are then adaptively integrated with spatial-domain semantic features to improve detail preservation while suppressing clutter-induced responses. Finally, the compensated shallow details are propagated into multiscale detection features using detail-preserving downsampling and efficient contextual enhancement. Extensive experiments on multiple remote sensing benchmarks, including optical, infrared, and synthetic aperture radar (SAR) datasets, demonstrate that WE-DETR achieves competitive performance compared with state-of-the-art detectors while maintaining a favorable balance between accuracy and computational cost. These results validate the effectiveness of the proposed wavelet-enhanced multiscale feature compensation framework across representative remote sensing small object detection tasks. The code is available at https://github.com/jingmingliang/WE-DETR**

***Index Terms*—Detection transformer, multiscale feature compensation, small object detection, wavelet-enhanced.**

I. Introduction

SMALL object detection in remote sensing imagery has attracted increasing attention due to its importance in applications such as traffic surveillance, maritime monitoring, military reconnaissance, and urban management. However, detecting small targets remains highly challenging because they occupy only a limited number of pixels and exhibit weak semantic cues, making them easily overwhelmed by complex backgrounds. Moreover, the multistage downsampling operations commonly used in deep neural networks inevitably lead to substantial loss of fine-grained details such as edges and textures, which are crucial for distinguishing small objects from surrounding clutter.

Recent advances in transformer-based remote sensing methods have demonstrated the importance of robust representation learning for complex scenes and small object detection. RIFormer+ [1] improves rotation-invariant feature learning through transformer-based representation modeling, enhancing the robustness of remote sensing feature learning at both long-overlooked local-level and the acknowledged global-level. FPNFormer [2] investigates the collaborative modeling of rotation-invariant and rotation-equivariant representations to better capture orientation-sensitive structures in remote sensing imagery. In hyperspectral remote sensing, group-aware hierarchical transformer [3] has also shown strong capability in modeling structured spectral–spatial dependencies through hierarchical feature interaction mechanisms. These studies suggest that effective representation learning is critical for improving the discriminability of remote sensing targets under complex imaging conditions. However, most existing transformer-based approaches primarily emphasize semantic representation enhancement and contextual interaction while paying limited attention to the preservation of fine-grained

Received 29 December 2025; revised 24 July 2026; accepted 31 August 2026. Date of publication 3 September 2026; date of current version 21 September 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 12571559, in part by the Key Research and Development Program of Henan Province under Grant 241111114000, in part by the Natural Science Foundation of Henan Province under Grant 262300420650, in part by the University-Enterprise Collaborative Innovation Project of Henan Province under Grant 26AXQXT006, in part by the Project of the Research and Development Center for College Mathematics Teaching in Higher Education Institutions under Grant CMC20240609, in part by the State Key Laboratory of Spatial Datum under Grant SKLSD2025-HNZZ-02, and in part by the Postgraduate Education Reform and Quality Improvement Project of Henan Province under Grant YJS2026ZYKC43. *(Yu Song and Weiguo Zuo are co-first authors.) (Corresponding authors: Xiaohui Yang; Zhenzhen Liu.)*

Yu Song is with the State Key Laboratory of Spatial Datum, College of Remote Sensing and Geoinformatics Engineering, Faculty of Geographical Science and Engineering, Henan University, Zhengzhou 450046, China, and also with the Henan Engineering Research Center for Artificial Intelligence Theory and Algorithms, School of Mathematics and Statistics, Henan University, Kaifeng 475002, China (e-mail: syu@henu.edu.cn).

Weiguo Zuo, Chen Zheng, and Xiaohui Yang are with Henan Engineering Research Center for Artificial Intelligence Theory and Algorithms, School of Mathematics and Statistics, Henan University, Kaifeng 475002, China (e-mail: wgzuo27@163.com; zhengchen_data@126.com; xhyang@henu.edu.cn).

Ronghua Shang is with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, School of Artificial Intelligence, Xidian University, Xi'an 710071, China (e-mail: rhshang@mail.xidian.edu.cn).

Haixiao Zhuang is with Beijing Institute of Spacecraft System Engineering, Beijing 100094, China (e-mail: zhx03@126.com).

Zhenzhen Liu is with the State Key Laboratory of Spatial Datum, College of Remote Sensing and Geoinformatics Engineering, Faculty of Geographical Science and Engineering, Henan University, Zhengzhou 450046, China (e-mail: liuzhenzhen@henu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2026.3730195

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:16:19 UTC from IEEE Xplore. Restrictions apply.

spatial details during feature extraction and downsampling. As a result, high-frequency (HF) structural information associated with tiny targets may still be progressively degraded in deep feature hierarchies.

From the perspective of feature preservation, many existing studies attempt to alleviate small object degradation by enhancing shallow high-resolution representations or redesigning downsampling operators. Multiscale architectures such as FPN and PANet strengthen the propagation of shallow features to deeper layers through top–down and lateral connections, thereby improving the detectability of small objects. High-resolution backbones such as HRNet [4] maintain parallel multiscale branches to reduce information loss during repeated downsampling. In parallel, improved downsampling operators such as antialiasing downsampling [5] and UniRepLKNet [6] have been proposed to preserve richer spatial details while suppressing aliasing effects. Wavelet-based operators, including wavelet pooling and discrete wavelet transform (DWT)-based downsampling [7], [8], further demonstrate that frequency-preserving transforms can effectively retain texture cues that are typically discarded in conventional CNN pipelines.

These studies indicate that preserving structural and frequency-related information is beneficial for remote sensing small object detection. Despite these advances, these methods still operate within conventional backbone structures, where early downsampling remains mandatory. Once structurally meaningful HF responses are suppressed during early feature extraction, later modules can only enhance incomplete representations rather than recover the originally lost details. As a result, existing feature fusion or spatial-preserving operators primarily strengthen already-degraded features rather than compensating for the originally lost details. This limitation is particularly detrimental to remote sensing small object detection, where discriminative cues often correspond to only a few edge, contour, or texture pixels. Although various feature-enhancement and detail-preserving strategies have been explored, existing methods still lack a unified mechanism for compensating HF structural information throughout the detection pipeline. The key challenge is to establish a coherent compensation mechanism for HF structural information throughout hierarchical feature extraction. Such a mechanism should preserve structurally meaningful details before severe downsampling-induced degradation occurs, selectively enhance remote sensing-specific structural responses, suppress clutter-induced or modality-specific noise, and effectively propagate compensated information into multiscale detection features.

To address these limitations, we propose WE-DETR, a wavelet-enhanced multiscale feature compensation detection transformer. WE-DETR builds a remote sensing-oriented progressive HF feature compensation mechanism. This mechanism aims to preserve fine-grained structural cues before they are irreversibly degraded by early downsampling, enhance direction-sensitive HF components under low-frequency structural guidance, and fuse the compensated frequency-domain cues with spatial-domain semantic representations to suppress unreliable background responses. In particular, WE-DETR introduces three remote sensing-specific design choices. First, wavelet decomposition is performed at the early feature extraction stage rather than after deep feature abstraction, so that edge, contour, and texture cues of small targets can be compensated before severe resolution degradation occurs. Second, different from conventional wavelet-based enhancement that treats subbands uniformly, the low-frequency subband is used to generate orientation-aware structural priors, which guide the enhancement of horizontal, vertical, and diagonal HF subbands. This design is particularly suitable for remote sensing targets with clear geometric boundaries and direction-dependent structures. Third, the enhanced wavelet-domain features are not simply concatenated with backbone features. Instead, they are adaptively fused with spatial-domain semantic features, allowing useful small object details to be strengthened while reducing the risk of amplifying clutter-induced HF noise.

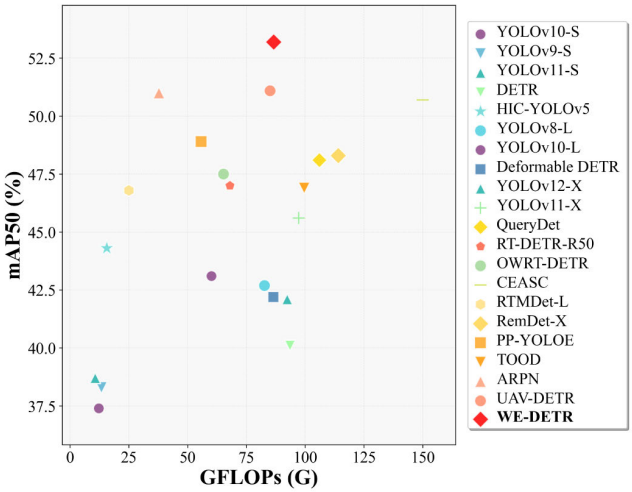


Fig. 1. Performance comparison of mainstream object detectors on VisDrone dataset (mAP_{50} versus GFLOPs).

To further strengthen multiscale representations, we design a multiscale enhanced feature fusion network (MSEF-Net) that preserves fine-grained details and enriches contextual modeling at critical feature levels. In particular, SPDConv and CSP-based decomposed large-kernel attention (DLKA) are used as supporting components within the compensation pathway. SPDConv helps propagate the compensated shallow HF details into the key detection scale without standard downsampling-induced information loss, while the CSP-based DLKA provides efficient contextual modeling for the fused multiscale features. In this way, WE-DETR forms a coherent compensation process from early frequency preservation to multiscale semantic detection.

Extensive experiments conducted on multiple remote sensing detection benchmarks demonstrate that WE-DETR achieves significant improvements in small object detection performance over state-of-the-art methods, highlighting the effectiveness of integrating wavelet-domain information and the proposed multiscale feature compensation strategy. Fig. 1 shows the comparison of WE-DETR with mainstream methods on the VisDrone dataset in terms of accuracy and computational complexity.

The main contributions of this work are summarized as follows.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:16:19 UTC from IEEE Xplore. Restrictions apply.

1) A wavelet-enhanced multiscale feature compensation detection transformer is proposed to preserve, enhance, fuse, and propagate fine-grained structural information before it is degraded by hierarchical feature transformation, addressing the detail loss problem of small object detection in remote sensing imagery.
2) A low-frequency-guided directional HF enhancement strategy is introduced to effectively exploit frequency-domain information. Instead of treating wavelet subbands uniformly, the low-frequency component is used to generate orientation-aware structural priors that selectively guide HF subbands.
3) A frequency–spatial multiscale compensation framework is developed to integrate enhanced wavelet features, shallow details, and semantic representations.
4) Extensive experiments on optical, infrared, and synthetic aperture radar (SAR) remote sensing benchmarks demonstrate that WE-DETR achieves superior small object detection performance while maintaining competitive computational efficiency.

II. Related Work

A. General Object Detection

In recent years, deep-learning-based object detection algorithms have achieved remarkable progress and can be broadly categorized into two main paradigms: two-stage and one-stage detectors. Representative two-stage approaches, such as R-CNN [9], fast R-CNN [10], and faster R-CNN [11], first generate region proposals using a region proposal network (RPN) and subsequently perform classification and bounding-box regression on these proposals. Although these methods typically deliver high detection accuracy, their multistep pipeline results in relatively slow inference speed.

On the contrary, one-stage detectors, such as SSD [12] and the YOLO family [13], directly predict object categories and bounding boxes on dense feature maps without the proposal generation step, enabling much faster inference. However, most of these CNN-based methods inherently rely on local receptive fields, making it difficult to effectively model long-range dependencies between objects and their surrounding context.

With the success of Transformers in computer vision, DETR [14] and its variants have been introduced into object detection. Among them, RT-DETR [15] stands out by employing an efficient hybrid encoder and a decoupled query selection mechanism, achieving the first real-time detection framework within the DETR family. Notably, it surpasses many CNN-based detectors in both accuracy and speed. Despite their strong performance in general scenes, these detectors are primarily designed for natural images. When directly applied to remote sensing imagery, they often struggle to cope with severe scale variations and complex background clutter.

B. Small Object Detection in Remote Sensing Imagery

Small object detection in remote sensing imagery poses unique challenges compared with generic object detection in natural images. Targets often occupy only a few tens of pixels, while complex and cluttered backgrounds, large variations in scale and orientation, and dense object distributions further complicate reliable detection. To address these challenges, a growing body of research in remote sensing has developed specialized techniques, which can be broadly categorized into three complementary directions: refined label and metric design, enhanced feature representation and multiscale fusion, and architecture-level adaptations for tiny-object detection.

Representative works along these directions have demonstrated notable improvements. Zhou and Zhu [16] replace conventional IoU- or center-based assignment strategies with 2-D Gaussian modeling and a Kullback–Leibler divergence (KLD)-based localization metric, allowing nonoverlapping tiny objects to be treated as valid positive samples and improving both classification and localization accuracy. Meanwhile, Ma et al. [17] propose an efficient feature-enhanced real-time detection network based on a Transformer architecture, which integrates a cross-scale feature pyramid interaction module (CFPIM), multiscale sensing fusion (MSSF), and a small object enhancement module (SOEM) to strengthen weak object features, fuse multiscale contextual information, and suppress background interference. Beyond these approaches, Zhang et al. [18] introduce a hierarchical coarse-to-fine feature learning strategy for salient maritime ship detection, which effectively combines global contextual modeling with local feature refinement to improve target discrimination in complex optical remote sensing scenes. Song et al. [1] propose RIFormer+, which rethinks rotation-invariant feature learning in Transformer architectures by enhancing rotation robustness through improved representation modeling, thereby improving the adaptability of remote sensing feature extraction under diverse target orientations. In addition, Zhang et al. [19] introduce knowledge distillation and efficient network design for aerial person detection, achieving a favorable tradeoff between detection accuracy and computational cost for edge deployment scenarios.

Beyond loss and detector-level improvements, more recent works address inherent feature representation limitations. For instance, Liu et al. [20] introduce a denoising-FPN that suppresses noise during multiscale feature fusion, combined with Transformer-based detection to better capture tiny-object representations under complex backgrounds. Similarly, Jiang et al. [21] adopt a multistage feature fusion strategy with saliency modeling, multidimensional attention, and a progressive detection head. Overall, recent studies consistently highlight that the main difficulties in small object detection arise from the loss of HF and spatial-detail information during downsampling. This reinforces the motivation of our work: instead of relying solely on spatial-domain multiscale designs, we introduce an explicit frequency-domain compensation module to preserve and enhance critical fine-grained cues for small objects.

C. Frequency-Domain Analysis in Deep Learning

Recent studies have increasingly incorporated frequency-domain representations into deep neural networks to preserve HF structures and mitigate information degradation caused

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:16:19 UTC from IEEE Xplore. Restrictions apply.

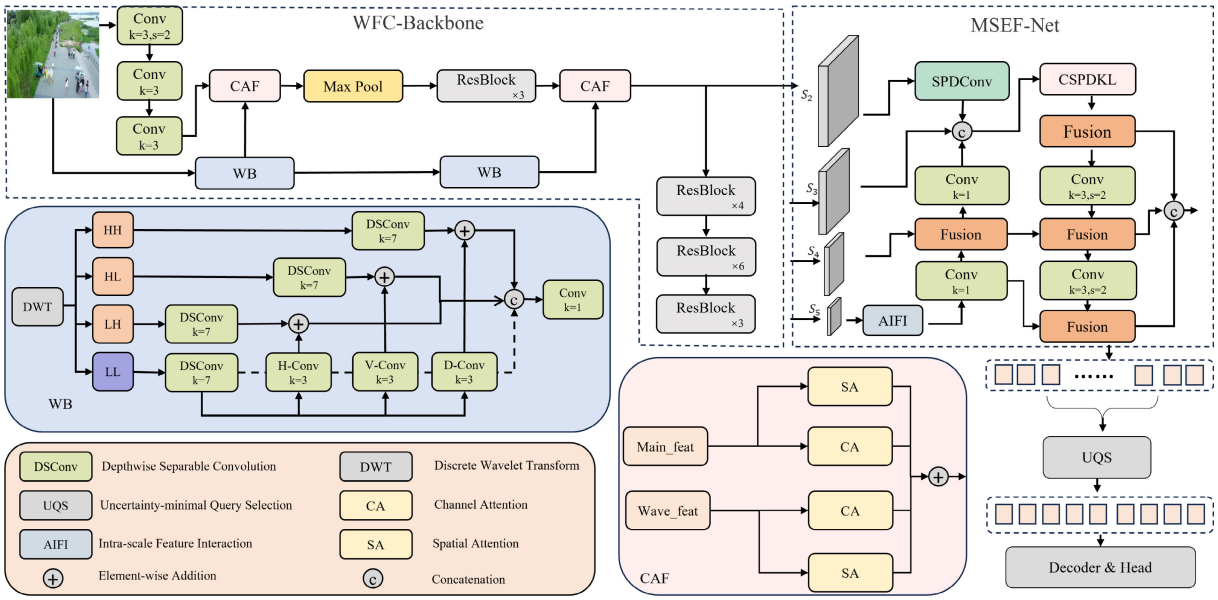


Fig. 2. WE-DETR architecture comprises three core components: 1) WFC-Backbone, which collaboratively extracts spatial and frequency-domain features through parallel branches; 2) multiscale enhancement fusion network (MSEF-Net), utilizing SPDConv and CSPDKL modules to efficiently integrate hierarchical features; and 3) standard DETR decoder and prediction head for final object localization and classification.

by spatial downsampling. Among these approaches, wavelet-based modeling has emerged as an effective means of decomposing features into multiband components with explicit spatial–frequency localization. For example, Finder et al. [22] develop WTConv, which embeds wavelet-filtered convolution kernels to enlarge the receptive field and maintain multifrequency responses, demonstrating strong advantages for fine-detail preservation.

Wavelet-guided cross-domain fusion has also been explored to bridge spatial and frequency cues. Recent works employ wavelet attention, wavelet pooling, or multiband enhancement modules to strengthen HF responses for detection, segmentation, and remote sensing understanding [23], [24]. These designs consistently demonstrate that explicitly modeling HF subbands improves the robustness of deep models against aliasing and enhances object-boundary precision. Furthermore, multilevel DWT embeddings have been incorporated into encoder–decoder networks for medical image segmentation and change detection, where preserving fine structural details is critical [25].

Overall, these studies confirm that explicit multiband decomposition in the frequency domain is a strong complement to spatial-domain learning, providing a principled mechanism for retaining HF cues that are easily degraded by convolutional or attention-based downsampling. This motivates our design: integrating a wavelet-based HF compensation branch with cross-domain attention enables the detector to preserve structurally discriminative HF signals that are especially crucial for small object recognition.

III. Methodology

A. Overall Architecture

To address the challenges of small object detection—such as variable object scales, complex backgrounds, and rich texture details—we developed an innovative and efficient hybrid network (WE-DETR). The core concept of this network is to achieve deep collaboration between spatial and frequency-domain features while effectively integrating multiscale information. As shown in Fig. 2, its architecture consists of three key components: 1) a wavelet-enhanced feature compensation backbone (WFC-Backbone); 2) a multiscale enhancement fusion network (MSEF-Net); and 3) a standard DETR decoder and prediction head for final object localization and classification. Our primary innovation lies in the collaborative design between the backbone network and the encoder.

B. Wavelet-Enhanced Feature Compensation Backbone

HF details, such as object boundaries, corners, and fine contours, are essential for detecting tiny targets whose discriminative cues occupy only a few pixels. Traditional CNN backbones inevitably weaken these HF structures once strided convolutions are applied. To counteract this degradation, we introduce a WFC-Backbone to explicitly preserve HF structures and to enhance their interaction with spatial-domain semantic features. As illustrated in Fig. 2, it consists of two parallel branches: 1) a wavelet branch (WB) that decomposes the input into multiband frequency subcomponents and performs direction-aware enhancement, and 2) a cross-attention fusion (CAF) module that enables bidirectional interaction between spatial and frequency representations.

1) Wavelet Branch: To counteract this degradation, we introduce a WB that separates the input into directional HF subbands, allowing the model to retain and enhance structural cues from the earliest stages.

Given an input feature map $X \in \mathbb{R}^{C\times H\times W}$, a single-level DWT produces four subbands

$$\{X_{LL}, X_{LH}, X_{HL}, X_{HH}\} = \mathrm{DWT}(X) \tag{1}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:16:19 UTC from IEEE Xplore. Restrictions apply.

where X_{LL} captures low-frequency approximations and $\{X_{LH}, X_{HL}, X_{HH}\}$ represents horizontal, vertical, and diagonal HF components, respectively.

Instead of treating the four subbands equally, we leverage the low-frequency component as a source of coarse structural guidance. In particular, X_{LL} is first processed by a depthwise separable convolution to extract contextual representations

$$F_{LL} = \phi_{DSC}(X_{LL}) \tag{2}$$

where ϕ_{DSC} denotes a depthwise separable convolution–normalization–activation block. Then, to capture orientation-aware structural priors, F_{LL} is further processed to extract three directional responses along the horizontal, vertical, and diagonal directions using fixed orientation-selective filters G_h, G_v, and G_d, respectively. These responses provide coarse structural priors that guide the enhancement of corresponding HF subbands. Each HF subband is then modulated by the matching directional prior, producing enhanced frequency features

$$\begin{aligned} F_{LH} &= \phi_{DSC}(X_{LH}) + G_h(F_{LL}) \\ F_{HL} &= \phi_{DSC}(X_{HL}) + G_v(F_{LL}) \\ F_{HH} &= \phi_{DSC}(X_{HH}) + G_d(F_{LL}). \end{aligned} \tag{3}$$

This operation enforces orientation consistency between low-frequency and HF components, enabling clearer representation of edges, corners, and fine textures that are characteristic of man-made structures in remote sensing imagery.

Finally, the enriched HF representations and the contextualized low-frequency feature are concatenated along the channel dimension and projected through a lightweight fusion layer. This yields

$$F_{wave} = \phi_{fus}([F_{LL}; F_{LH}; F_{HL}; F_{HH}]) \tag{4}$$

where ϕ_{fus} denotes a normalization–activation–projection block.

Through the joint modeling of multilevel DWT and WB, WFC-Backbone maintains the convolutional backbone structure while introducing a direction-sensitive frequency-domain prior to early features. This enables the network to exhibit enhanced edge and detail perception when processing high-density small targets and complex textured backgrounds.

2) CAF Module: Although the WB explicitly preserves directional HF structures, frequency-domain features alone lack sufficient semantic abstraction for robust object discrimination. Conversely, spatial-domain backbone features provide rich semantic representations but tend to suppress fine structural details during hierarchical downsampling. Simple concatenation of these two representations is therefore suboptimal, as it fails to explicitly model their complementary roles. To address this limitation, we introduce a CAF module that enables bidirectional interaction between spatial-domain semantic features and frequency-domain structural features.

Let F_{main} and F_{wave} denote the spatial-domain backbone feature and the frequency-domain feature generated by the WB, respectively. The objective of CAF is to jointly emphasize semantically meaningful channels and spatially salient regions across both domains.

Channel-wise attention is first applied to adaptively recalibrate the importance of different feature channels. For each domain, global average pooling (GAP) is employed to aggregate spatial information into a channel descriptor

$$z_{main} = \mathrm{GAP}(F_{main}), \quad z_{wave} = \mathrm{GAP}(F_{wave}). \tag{5}$$

The descriptors are then projected through a shared lightweight transformation to generate channel attention weights

$$a^c_{main} = \sigma(\phi_c(z_{main})), \quad a^c_{wave} = \sigma(\phi_c(z_{wave})) \tag{6}$$

where $\phi_c(\cdot)$ denotes a bottleneck channel projection implemented using fully connected layers, and $\sigma(\cdot)$ is the sigmoid activation function. The channel-refined features are obtained as follows:

$$F^c_{main} = F_{main} \odot a^c_{main}, \quad F^c_{wave} = F_{wave} \odot a^c_{wave} \tag{7}$$

where $\odot$ denotes element-wise multiplication. This operation highlights discriminative semantic channels in the spatial domain while emphasizing structurally informative channels in the frequency domain.

To further localize salient regions corresponding to small objects, spatial attention (SA) is applied to both domains. For each feature map, channel-wise aggregation is performed using average pooling and max pooling

$$\begin{aligned} s_{main} &= \left[\mathrm{AvgPool}(F_{main}); \mathrm{MaxPool}(F_{main})\right] \\ s_{wave} &= \left[\mathrm{AvgPool}(F_{wave}); \mathrm{MaxPool}(F_{wave})\right]. \end{aligned} \tag{8}$$

The aggregated representations are then passed through a shared spatial projection to generate SA maps

$$a^s_{main} = \sigma(\phi_s(s_{main})), \quad a^s_{wave} = \sigma(\phi_s(s_{wave})) \tag{9}$$

where $\phi_s(\cdot)$ is implemented using a 7×7 convolution. The spatially refined features are computed as follows:

$$F^s_{main} = F_{main} \odot \mathbf{a}^s_{\mathrm{main}}, \quad F^s_{wave} = F_{wave} \odot \mathbf{a}^s_{\mathrm{wave}}. \tag{10}$$

This mechanism enhances spatial regions that are consistently activated in either semantic or structural representations while suppressing background clutter.

Finally, the channel-refined and spatial-refined features from both domains are aggregated to form the fused representation

$$F_{\mathrm{CAF}} = F^c_{main} + F^s_{main} + F^c_{wave} + F^s_{wave}. \tag{11}$$

This additive fusion strategy enables mutual reinforcement between spatial-domain semantics and frequency-domain structural cues. HF information guides the activation of small object regions that may be weak in semantic features, while semantic context suppresses spurious HF responses caused by complex backgrounds or noise.

Through this cross-attention mechanism, CAF establishes an explicit interaction between spatial-domain and frequency-domain representations, yielding structurally aware and semantically enriched features that are better suited for small object detection in remote sensing imagery.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:16:19 UTC from IEEE Xplore. Restrictions apply.

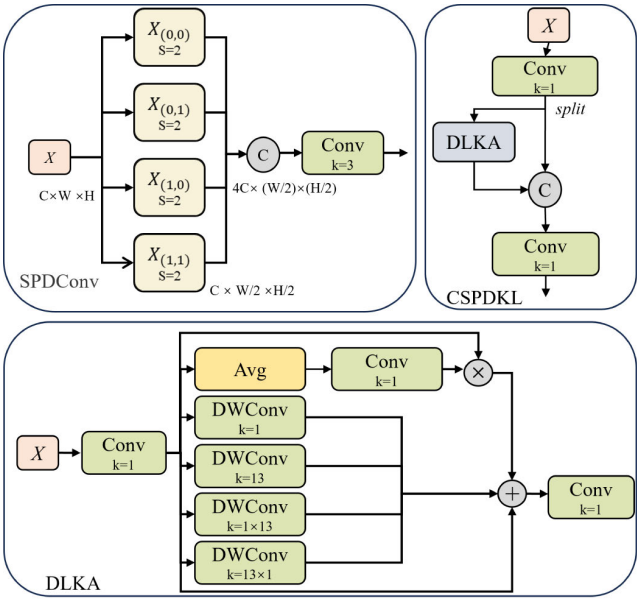


Fig. 3. Architectural overview of the key components in MSEF-Net, including the detail-preserving downsampling module SPDConv and the context enhancement block CSPDKL for multilevel feature aggregation.

C. Multiscale Enhanced Feature Fusion Network

The multiscale enhanced feature fusion network (MSEF-Net) is designed to address two critical challenges in small object detection within remote sensing imagery: 1) the severe loss of fine-grained information during early downsampling and 2) the insufficient contextual modeling at the key detection scale where most small targets appear. To overcome these issues, MSEF-Net integrates a detail-preserving downsampling strategy with a computationally efficient large-kernel attention mechanism, enabling effective multilevel feature aggregation while retaining crucial spatial details. Fig. 3 provides detailed illustrations of the key components.

In conventional CNN backbones, strided convolutions reduce the spatial resolution of feature maps at an early stage, inevitably discarding pixel-level details that are vital for representing small objects. To mitigate this, we adopt a detail-preserving downsampling module based on SPDConv [26], which replaces strided convolution with a space-to-depth rearrangement. In particular, the shallow feature map $\mathbf{S}_2$ is partitioned into nonoverlapping spatial blocks that are reorganized along the channel dimension, allowing all pixel information to be preserved even as the spatial resolution is halved. A lightweight 3×3 convolution is then applied to encode local patterns while compressing the expanded channel dimension. This produces a transformed feature $\widetilde{\mathbf{S}}_2$ that contains rich HF cues originally present in the shallow layer but aligned to the same spatial scale as deeper features. This operation has proven particularly beneficial for small object detection, as it avoids the information loss typically introduced by standard downsampling and maintains structural details such as edges, textures, and fine contours.

After obtaining the detail-preserved shallow feature $\widetilde{\mathbf{S}}_2$, MSEF-Net performs multisource fusion at the $\mathbf{S}_3$ stage, where complementary information from different depths can be effectively integrated. In particular, three types of features are aligned and aggregated at this resolution: the shallow feature $\widetilde{\mathbf{S}}_2$ containing HF textures and fine contours, the intermediate semantic feature $\mathbf{S}_3$, and the deeper contextual feature $\widetilde{\mathbf{S}}_4$, which is produced by fusing $\mathbf{S}_4$ and $\mathbf{S}_5$ and subsequently upsampling it to match the spatial size of $\mathbf{S}_3$. These three components, respectively, carry detailed local cues, midlevel structural semantics, and high-level contextual information. They are concatenated along the channel dimension to form a unified multiscale representation, enabling the network to benefit simultaneously from fine-grained shallow features and broad contextual cues from deeper layers, thus establishing a more expressive and comprehensive feature basis for subsequent enhancement.

Algorithm 1 Procedure of the Proposed MSEF-Net

Input: Backbone feature maps $\mathbf{S}_2, \mathbf{S}_3, \mathbf{S}_4, \mathbf{S}_5$
Output: Enhanced multi-scale feature map F_{MSEF}
Detail-Preserving Downsampling
$\widetilde{\mathbf{S}}_2 \leftarrow \text{SPD-Conv}(\mathbf{S}_2)$
Deep Context Aggregation
$\widetilde{\mathbf{S}}_4 \leftarrow \text{Upsample}(\text{ContextFusion}(\mathbf{S}_4, \mathbf{S}_5))$
Multi-Source Feature Fusion
$F_{\text{cat}} \leftarrow \text{Concat}(\widetilde{\mathbf{S}}_2, \mathbf{S}_3, \widetilde{\mathbf{S}}_4)$
Context Enhancement
$[F_a, F_b] \leftarrow \text{ChannelSplit}(F_{\text{cat}})$
$F'_a \leftarrow \text{DLKA}(F_a)$
$F_{\text{MSEF}} \leftarrow \text{Projection}(F'_a, F_b)$
return F_{MSEF}

To enhance long-range contextual modeling on the fused feature $F_{\mathbf{cat}}$, we introduce a DLKA operator that captures directional spatial dependencies. In particular, DLKA consists of three parallel depthwise convolutions designed to model horizontal, vertical, and 2-D spatial interactions, respectively. This decomposition enables efficient extraction of direction-aware responses, which is particularly beneficial for modeling elongated structures and anisotropic patterns commonly observed in remote sensing imagery. To further improve computational efficiency and facilitate feature reuse, DLKA is embedded into a CSP-style structure, where only a subset of channels is processed by DLKA while the remaining channels are preserved through a shortcut branch. We refer to this CSP-style integration of DLKA as CSPDKL, which serves as the context enhancement block in MSEF-Net. Algorithm 1 summarizes the forward feature aggregation process of the proposed MSEF-Net.

Through this combination of detail-preserving downsampling and large-kernel contextual modeling, MSEF-Net enables a balanced integration of shallow HF cues, midlevel semantic structures, and deep contextual signals. The preserved shallow textures improve the visibility of small objects, the fused multiscale features at $\mathbf{S}_3$ consolidate complementary information, and the CSPDKL block provides broad contextual understanding without high computational cost. Together, these components ensure that the resulting representations are both structurally rich and semantically discriminative, greatly

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:16:19 UTC from IEEE Xplore. Restrictions apply.

enhancing the model's ability to detect tiny targets in complex remote sensing scenarios.

IV. Experiments and Discussions

In this section, to validate the proposed method, we conducted comprehensive experimental evaluations on widely used datasets including VisDrone [27], NWPU VHR-10 [28], HIT-UAV [29], and SSDD. We also conducted comprehensive ablation studies to thoroughly evaluate our contributions.

A. Datasets and Metrics

1) Datasets: The VisDrone dataset is currently one of the most influential large-scale benchmark datasets in the field of UAV computer vision. The dataset contains 10 209 still images (6471 for training, 548 for validation, and 3190 for testing), with a total of ten object categories.

The NWPU VHR-10 dataset, released by the China Meteorological Administration in 2014, contains 800 high-resolution satellite images, including 650 target images and 150 background images. These images were sourced from Google Earth and the Vaihingen dataset, cropped, and manually annotated by experts. The dataset is divided into three subsets: training (454 images), validation (32 images), and testing (164 images).

The HIT-UAV dataset captures infrared thermal images from diverse environments including schools, parking lots, roads, and playgrounds. It contains 2898 infrared thermal images with varying flight altitudes, camera angles, and lighting conditions, including person (PE), car (CA), bicycle (BI), other vehicle (OT), and do not care (DO). The dataset is divided into three sets: 2029 images for training, 290 images for validation, and 579 images for testing.

SSDD is a dataset specifically designed for ship target detection in SAR imagery. The dataset comprises 1160 SAR images with a total of 2456 ship targets. The ships in the dataset exhibit various scales and orientations against complex and diverse backgrounds, including open sea, inshore areas, and harbor scenes, presenting significant challenges for detection algorithms. The dataset contains a single object category: ships. Following a 7:2:1 split ratio, the dataset is partitioned into training (807 images), validation (222 images), and testing (131 images) subsets.

2) Metrics: In accordance with standard evaluation protocols in computer vision, we employ several metrics to measure the precision of object detection. The mean average precision (mAP) and mAP_{50} are used to measure the object detection performance. mAP is calculated by averaging the AP values at multiple IoU thresholds ranging from 0.5 to 0.95 with a stride of 0.05. mAP_{50} denotes the detection mAP with IoU = 0.5. To evaluate model efficiency, we use the number of parameters [Params(M)] and FLOPs(G) to estimate the computational complexity.

B. Implementation Details

All experiments are conducted under a unified hardware and software environment to ensure reproducibility and fair comparison. The training is performed on a workstation equipped with two NVIDIA A100 GPUs. Our model is trained for a total of 130 epochs. To facilitate more stable convergence, we adopt a differential learning rate strategy: the backbone network is optimized with a smaller learning rate of 1×10^{-4}, while the other components use an initial learning rate of 1×10^{-3}. Unless otherwise specified, all other hyperparameters follow those used in RT-DETR.

C. Performance Comparison With State-of-the-Art Methods

1) VisDrone: We compare the proposed WE-DETR with a comprehensive set of state-of-the-art detectors on the VisDrone dataset, including mainstream YOLO-series models, representative DETR-based frameworks, and a number of methods specifically designed for small object detection in aerial imagery. As summarized in Table I, WE-DETR achieves an mAP of 31.9 and an mAP_{50} of 53.2, surpassing other advanced methods.

Compared with RT-DETR-R50, which is our baseline detector, WE-DETR achieves a clear performance gain of +1.6 mAP and +2.4 mAP_{50}. Moreover, WE-DETR shows strong competitiveness even when compared with models that possess substantially larger parameter scales. For instance, relative to YOLOv11-X and YOLOv12-X, WE-DETR achieves +3.8% and +4.2% mAP improvements while using only 76.5% and 73.3% of their parameter sizes, respectively. When compared with the small object detection RemDet-X, whose parameter count is approximately 1.7× that of WE-DETR, our method still attains nearly 2.0% higher mAP and surpasses it by 4.9% in mAP_{50}. Moreover, compared to BAF-Net, which contains nearly three times more parameters than WE-DETR, our model still achieves a 1.1% higher mAP and outperforms it by 0.6% in mAP_{50}. These observations highlight the superior localization accuracy and detection robustness achieved by WE-DETR. These comparisons indicate that the proposed architecture achieves a superior balance between accuracy and computational efficiency.

2) NWPU VHR-10: To further assess the effectiveness of WE-DETR in diverse remote sensing scenarios, additional experiments are conducted on the NWPU VHR-10 dataset. This dataset consists of high-resolution satellite imagery with large variations in object scale and complex background clutter, making it a widely recognized benchmark for evaluating the comprehensive performance of remote sensing object detectors. As summarized in Table II, WE-DETR is compared against 13 state-of-the-art detection methods.

The experimental results demonstrate that WE-DETR achieves highly competitive accuracy on this challenging dataset. In particular, WE-DETR attains an mAP of 64.2% and an mAP_{50} of 93.2%, outperforming both large-scale detectors and advanced methods tailored for small object detection in aerial and satellite imagery. Despite operating with a moderate model size and computational load, WE-DETR consistently delivers strong detection accuracy, indicating a favorable balance between precision and efficiency. These findings further verify that the proposed feature enhancement and architectural design enable WE-DETR to maintain robust performance across heterogeneous remote sensing environments.

3) HIT-UAV: To further evaluate the applicability of WE-DETR, additional experiments are conducted on the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:16:19 UTC from IEEE Xplore. Restrictions apply.

TABLE I

COMPARISON OF PERFORMANCE OF STATE-OF-THE-ART OBJECT DETECTORS ON THE VISDRONE DATASET

Model	Publication	Input Size	Params (M)	GFLOPs	mAP	mAP_{50}
YOLOv10-S [30]	NeurIPS2024	640×640	8.0	12.3	22.2	37.4
YOLOv9-S [31]	ECCV2024	640×640	7.2	13.4	22.7	38.3
YOLOv11-S [32]	arXiv2024	640×640	9.4	10.7	23.0	38.7
DETR [14]	ECCV2020	1333×750	60.0	93.5	24.1	40.1
HIC-YOLOv5 [33]	ICRA2024	640×640	9.4	15.6	26.0	44.3
YOLOv8-L	–	640×640	43.7	82.6	26.1	43.7
YOLOv10-L [30]	NeurIPS2024	640×640	24.4	60.2	26.3	43.1
Deformable DETR [34]	ICLR2020	1333×800	39.8	86.5	27.1	42.2
YOLOv12-X [35]	arXiv2025	640×640	59.3	92.3	27.7	42.1
YOLOv11-X [32]	arXiv2024	640×640	56.8	97.3	28.1	45.6
QueryDet [36]	CVPR2022	2400×2400	33.9	106	28.3	48.1
RT-DETR-R50 [15]	CVPR2024	640×640	40.9	67.9	30.3	50.8
OWRT-DETR [17]	TGRS2025	640×640	38.2	65.2	28.6	47.5
CEASC [37]	CVPR2023	1333×800	–	150	28.7	50.7
RTMDet-L [38]	CISAT2025	–	52.3	25.0	29.3	46.8
RemDet-X [39]	AAAI2025	640×640	74.1	114.0	29.9	48.0
PP-YOLOE-P2-Alpha-1 [40]	arXiv2022	640×640	54.1	55.7	30.1	48.9
TOOD [41]	ICCV2021	–	33.9	99.5	30.3	46.9
ARPN [42]	TCSVT2023	–	41.1	37.8	30.4	51.0
BAF-Net [43]	TGRS2025	1333×800	132.8	768.1	30.8	52.6
WE-DETR (OURS)	–	640×640	43.5	86.6	**31.9**	**53.2**

TABLE II

COMPARISON OF PERFORMANCE OF STATE-OF-THE-ART OBJECT DETECTORS ON THE NWPU VHR-10 DATASET

Model	Publication	Input Size	Params (M)	GFLOPs	mAP	mAP_{50}
YOLOv8-L	–	640×640	43.7	82.6	61.3	91.0
YOLOv8-X	–	640×640	68.2	128.9	61.6	91.9
YOLOv9-E [31]	ECCV2024	640×640	58.1	96.3	60.0	90.5
YOLOv10-L [30]	NeurIPS2024	640×640	24.4	63.5	58.7	91.3
YOLOv10-X [30]	NeurIPS2024	640×640	31.6	85.5	57.3	86.7
YOLOv11-X [32]	arXiv2024	640×640	56.9	97.4	59.7	92.7
YOLOv12-X [35]	arXiv2025	640×640	59.1	99.2	59.9	91.6
FFCA-YOLO [44]	TGRS2024	640×640	7.1	25.7	55.2	90.6
FPSOD [45]	TGRS2024	–	–	–	57.7	92.2
Mamba-YOLO-B [46]	AAAI2025	640×640	21.8	24.8	57.7	91.6
LMFF-MFEE [47]	TGARS2025	640×640	8.8	20.9	58.1	92.6
OWRT-DETR [17]	TGRS2025	640×640	38.2	65.2	58.9	92.3
RemDet-X [39]	AAAI2025	640×640	74.1	114.0	59.4	92.4
WE-DETR (OURS)	–	640×640	43.5	86.6	**64.2**	**93.2**

TABLE III

COMPARISON OF PER-CLASS AP (%) AND OVERALL MAP (%) OF STATE-OF-THE-ART OBJECT DETECTION METHODS ON THE HIT-UAV DATASET

Model	PE	CA	BI	OT	DO	mAP	Params (M)	GFLOPs
YOLOv8-L	74.6	52.2	55.5	59.9	46.3	57.7	43.7	82.6
YOLOv8-X	76.1	52.2	56.4	60.4	40.6	57.2	68.2	128.9
YOLOv9-E	75.8	52.5	56.6	**61.2**	43.4	57.9	58.1	96.4
YOLOv10-L	75.7	48.7	51.4	58.7	36.9	54.3	25.7	63.7
YOLOv10-X	75.1	49.4	53.2	58.4	38.5	54.9	31.6	85.6
YOLOv11-X	76.3	52.4	56.7	60.9	41.9	57.5	56.9	97.5
YOLOv12-X	**76.6**	51.1	54.6	60.6	44.2	57.4	59.1	99.5
RT-DETR-R50	74.5	51.8	53.0	58.7	45.0	56.6	40.9	67.9
CSFPR-RTDETR	73.4	51.3	55.6	56.8	45.4	56.5	14.1	63.9
OWRT-DETR	73.0	52.0	57.6	59.5	41.9	56.8	38.2	65.2
RemDet-X	73.2	52.4	53.7	58.9	40.9	55.8	74.1	114.0
WE-DETR (Ours)	74.0	**52.6**	**58.2**	60.3	**47.7**	**58.4**	43.5	86.6

HIT-UAV infrared small object dataset. As shown in Table III, WE-DETR is compared with 11 state-of-the-art detection methods and achieves the highest overall mAP of 58.4%, outperforming all competing approaches. Compared with the baseline RT-DETR-R50, WE-DETR improves the mAP from 56.6% to 58.4%, yielding a gain of 1.8 percentage points. In addition to the overall performance, WE-DETR achieves the best AP on three object categories (CA, BI, and DO),

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:16:19 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
COMPARISON OF PERFORMANCE OF STATE-OF-THE-ART OBJECT DETECTORS ON THE SSDD DATASET

Model	Params (M)	GFLOPs	mAP
YOLOv8-L	43.7	82.6	70.8
YOLOv8-X	68.2	128.9	71.0
YOLOv9-E	58.1	96.4	71.6
YOLOv10-L	25.7	63.7	70.6
YOLOv10-X	31.6	85.6	70.8
YOLOv11-X	56.9	97.5	71.6
YOLOv12-X	59.1	99.5	70.5
RT-DETR-R50	40.9	67.9	71.5
CSFPR-RTDETR	14.1	63.9	70.9
OWRT-DETR	38.2	65.2	71.7
RemDet-X	74.1	114.0	70.5
WE-DETR (OURS)	43.5	86.6	**72.7**

demonstrating its effectiveness in detecting different categories of infrared targets.

These results show that the proposed method is also effective on infrared remote sensing imagery, where target appearance is often affected by low contrast and background interference. The proposed wavelet-enhanced feature compensation strategy preserves fine-grained structural information during feature extraction while reducing the influence of background noise, leading to more discriminative feature representations for small targets. The experimental results indicate that the proposed design applies to infrared small object detection and consistently improves the baseline detector on the HIT-UAV dataset.

4) SSDD: To further evaluate the applicability of WE-DETR to SAR imagery, additional experiments are conducted on the SSDD dataset. As reported in Table IV, WE-DETR achieves an mAP of 72.7%, demonstrating competitive detection performance in complex SAR scenes characterized by speckle noise, background clutter, and large variations in target scale.

These results indicate that the proposed WE-DETR framework is also applicable to SAR-based ship detection. The wavelet-enhanced feature compensation strategy helps preserve informative structural details during feature extraction while reducing the influence of background interference, leading to more discriminative feature representations for ship detection in SAR imagery.

5) Visualization Performance: To further validate the detection performance of WE-DETR from a visual perspective, representative scenes from the natural-light VisDrone dataset, the infrared HIT-UAV dataset, and the SAR SSDD dataset are selected for qualitative comparison, as illustrated in Figs. 4–6. Fig. 4 presents four UAV images captured under high-density traffic and complex urban intersections. From left to right are the original images, followed by the detection results of OWRT-DETR, RemDet, and WE-DETR. It can be observed that WE-DETR is capable of detecting more true small-scale targets in distant regions and heavily occluded areas while simultaneously reducing false alarms.

Fig. 5 shows three infrared UAV images captured under varying altitudes and illumination levels. Similarly, from left to right are the original images, OWRT-DETR, RemDet, and WE-DETR. Compared with the two competing methods, WE-DETR maintains reliable detection performance even in low-contrast, noise-dominant regions, successfully localizing weak-texture small objects such as pedestrians and vehicles.

Fig. 6 shows representative SAR scenes with complex sea clutter and dense ship distributions. From left to right are the original SAR images, followed by the detection results of OWRT-DETR, RemDet, and WE-DETR. As illustrated, WE-DETR is able to more accurately localize ships with weak scattering responses and small spatial extents. In challenging near-shore scenes and densely packed ship regions, competing methods tend to produce missed detections, whereas WE-DETR maintains more complete and precise target coverage, especially for closely spaced and low-contrast ships.

The qualitative comparisons on VisDrone, HIT-UAV, and SSDD collectively demonstrate that WE-DETR delivers stable and accurate detection across different imaging modalities, flight altitudes, and scene complexities. These observations further demonstrate the effectiveness of the proposed framework across diverse remote sensing scenarios and indicate its potential for handling different imaging conditions.

D. Ablation Study

To evaluate the effectiveness and design appropriateness of WE-DETR, we conducted ablation experiments on the VisDrone2019 dataset. By progressively integrating these components, we analyze both the individual contributions of each component and their synergistic effects. In particular, we analyze whether frequency-domain information should be introduced at the early stage, whether plain wavelet decomposition is sufficient, whether direction-aware wavelet enhancement is necessary, and whether compensated shallow details should be further propagated and contextualized through multiscale feature fusion. The results are summarized in Table V, where each component is progressively introduced to verify its specific role in preserving, enhancing, fusing, and propagating fine-grained structural information.

1) Analysis of WFC-Backbone: We first investigate the design choices in the proposed WFC-Backbone. The original RT-DETR-R50 baseline, without DWT, WB, or CAF, obtains 30.3% mAP and 50.8% mAP_{50}. This setting corresponds to a conventional spatial-domain detection pipeline, where HF details such as edges, contours, and fine textures may be progressively weakened during early downsampling.

When plain DWT is introduced, the performance improves to 30.6% mAP and 51.2% mAP_{50}, yielding gains of +0.3% mAP and +0.4% mAP_{50} over the baseline. This result indicates that early frequency decomposition can provide useful complementary cues for small object detection. However, the improvement remains limited, suggesting that simply concatenating or injecting raw wavelet subbands is insufficient. This observation supports our motivation that wavelet decomposition itself is not the key contribution; rather, the frequency components need to be further organized and enhanced according to the structural characteristics of remote sensing targets.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:16:19 UTC from IEEE Xplore. Restrictions apply.

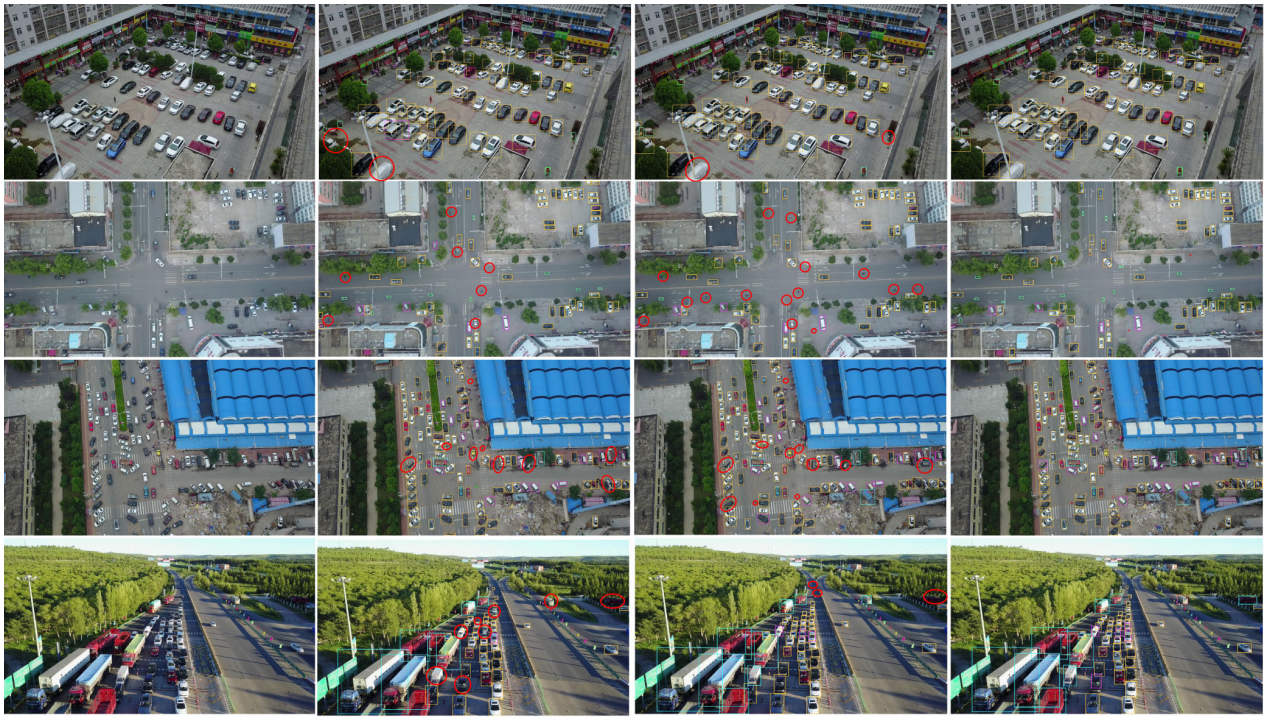


Fig. 4. Comparison of object detection results across different methods on the VisDrone dataset. Red circles highlight missed detections of the comparison models.

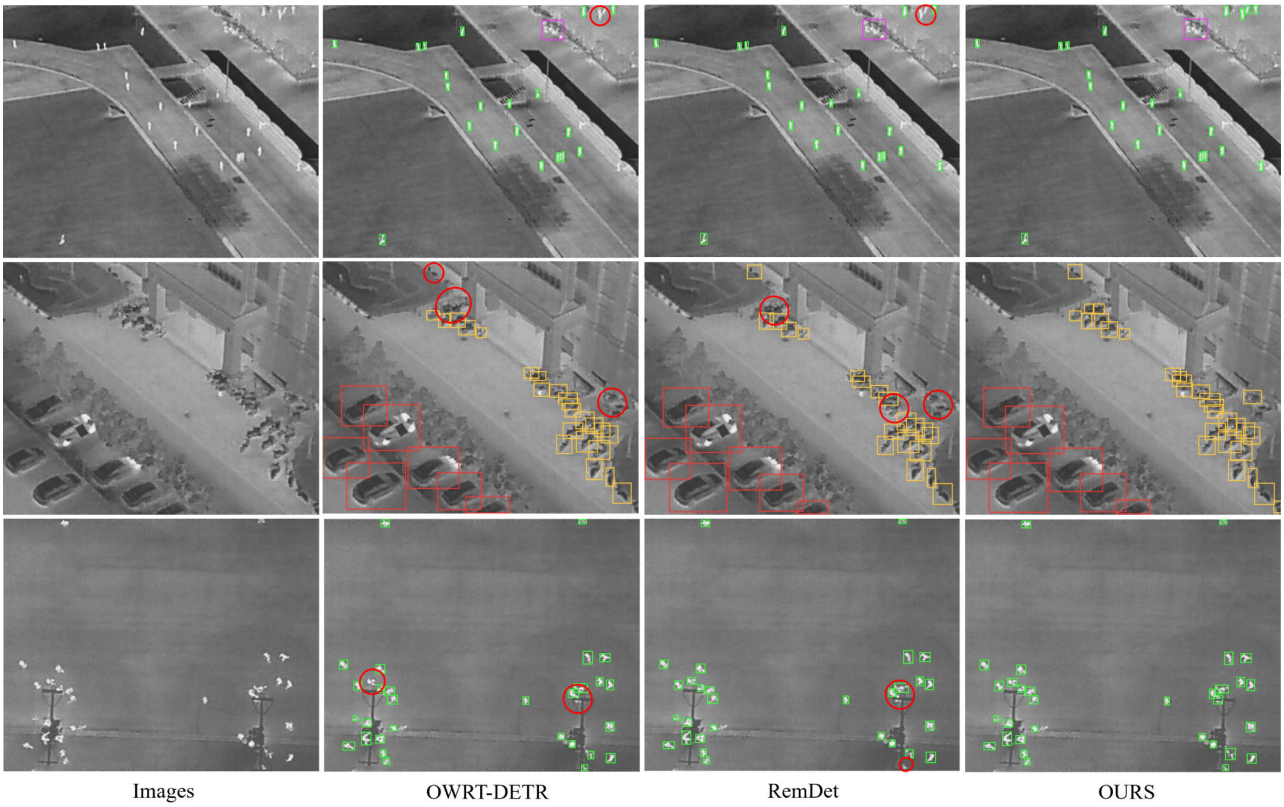


Fig. 5. Comparison of infrared target detection results across different methods on the HIT-UAV dataset. Red circles highlight missed detections of the comparison models.

After adding the proposed WB, the model achieves 30.8% mAP and 51.6% mAP_{50}. Compared with the plain DWT setting, this brings additional gains of +0.2% mAP and +0.3% mAP_{50}. This improvement demonstrates that the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:16:19 UTC from IEEE Xplore. Restrictions apply.

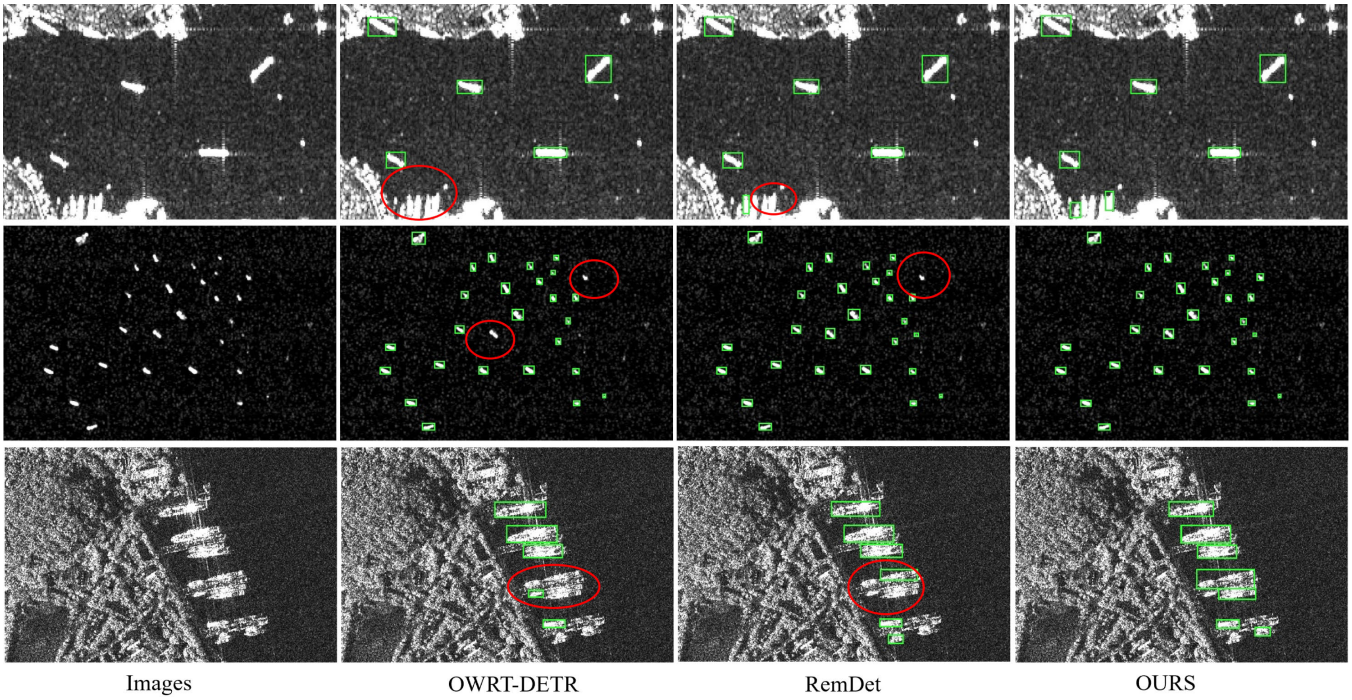


Fig. 6. Comparison of SAR target detection results across different methods on the SSDD dataset. Red circles highlight missed detections of the comparison models.

TABLE V
ABLATION STUDY OF DIFFERENT DESIGN CHOICES IN WE-DETR

DWT	WB	Wavelet Feature Fusion	S_2 Fusion	S_2 Propagation	CSPDKL	Params(M)	GFLOPs	mAP	mAP_{50}
×	×	–	×	–	×	40.89	67.92	30.3	50.8
✓	×	Add	×	–	×	40.93	68.13	30.6	51.2
✓	✓	Add	×	–	×	40.93	69.37	30.8	51.5
✓	✓	SA	×	–	×	40.96	69.30	30.9	51.6
✓	✓	CA	×	–	×	40.96	69.30	30.9	51.7
✓	✓	CAF	×	–	×	40.96	69.30	31.0	51.8
✓	✓	CAF	✓	Pooling	×	41.42	73.53	31.1	52.2
✓	✓	CAF	✓	SPDConv	×	42.53	80.66	31.4	52.5
✓	✓	CAF	✓	SPDConv	✓	43.47	86.59	31.9	53.2

proposed low-frequency-guided directional HF enhancement is more effective than using raw wavelet subbands directly. In particular, by using the low-frequency component to generate orientation-aware structural priors, the WB selectively strengthens horizontal, vertical, and diagonal HF responses. Such a design is well aligned with remote sensing imagery, where small targets such as vehicles, ships, and aircraft often exhibit clear edges, contours, and direction-dependent structures.

We further investigate different frequency–spatial fusion strategies for integrating the enhanced wavelet-domain features with spatial-domain semantic representations. Starting from direct feature addition, the model achieves 30.8% mAP and 51.5% mAP_{50}, demonstrating that wavelet features can provide complementary structural information for small object detection. However, simple addition treats all feature responses equally and lacks adaptive feature selection. By introducing SA or channel attention (CA), the performance is slightly improved to 30.9% mAP, with 51.6% and 51.7% mAP_{50}, respectively, indicating that selectively emphasizing spatial regions or informative channels is beneficial for frequency–spatial interaction. The proposed CAF further combines spatial and channel-wise feature recalibration, achieving 31.0% mAP and 51.8% mAP_{50}. This result verifies that spatial localization and channel dependency provide complementary information when fusing wavelet-domain details with spatial-domain semantics. Therefore, CAF is not merely an additional attention block but a dedicated fusion strategy that adaptively balances fine-grained structural compensation and semantic discrimination.

2) Analysis of MSEF-Net: We then evaluate the role of the multiscale enhanced feature fusion network. Based on the WFC-Backbone, we first introduce the $\mathbf{S}_2$ frequency–spatial fusion features with conventional pooling-based propagation. This setting improves the performance from 31.0% mAP and 51.8% mAP_{50} to 31.1% mAP and 52.2% mAP_{50}. The improvement in mAP_{50} indicates that shallow high-resolution representations provide more accurate localization cues for small objects. However, the limited gain in mAP suggests that directly propagating shallow features through conventional

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:16:19 UTC from IEEE Xplore. Restrictions apply.

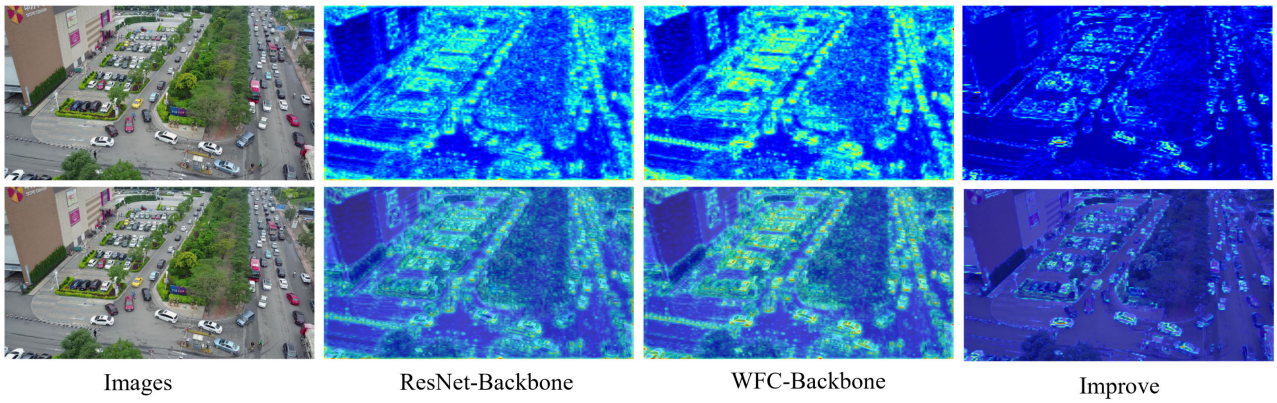


Fig. 7. Visualization of the feature activation maps produced by the baseline ResNet backbone and the proposed WFC-Backbone on representative VisDrone scenes containing densely distributed small objects. The first row presents the raw activation maps extracted from the backbone stage, and the second row overlays the activation maps onto the original images.

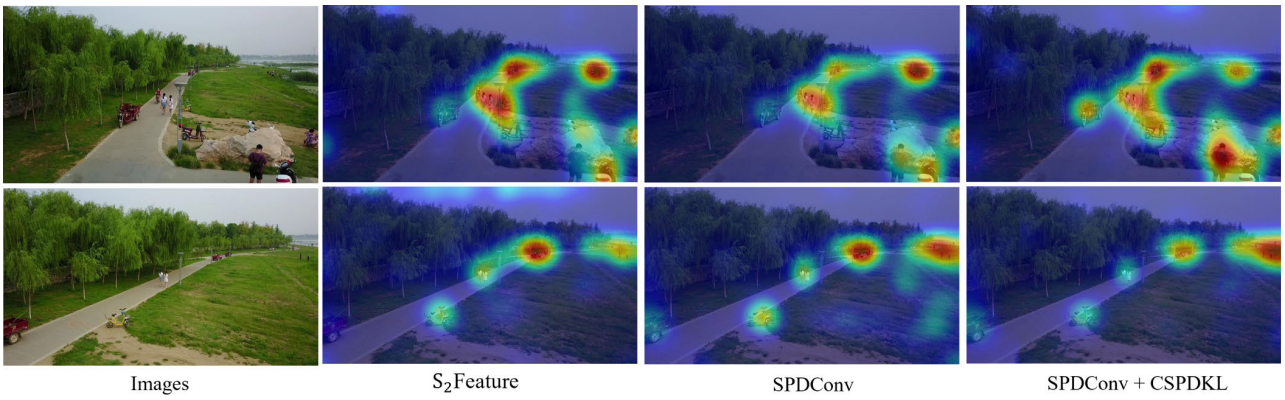


Fig. 8. Visualization of the attention heatmaps corresponding to different configurations of the proposed MSEF-Net.

downsampling may still lose part of their fine-grained structural information.

To further investigate how shallow details should be propagated, we replace the pooling-based downsampling with SPDConv for the $\mathbf{S}_2$ frequency–spatial fusion features. With SPDConv, the model achieves 31.4% mAP and 52.5% mAP_{50}, improving by +0.3% mAP and +0.3% mAP_{50} compared with the pooling-based propagation strategy. This result demonstrates that the propagation strategy is critical for exploiting compensated shallow features. Unlike conventional pooling-based downsampling, SPDConv preserves local spatial details during resolution reduction, allowing the frequency–spatial fusion features to maintain fine-grained structures while being effectively integrated into the multiscale detection hierarchy.

Finally, the CSP-based decomposed large-kernel context enhancement block is incorporated, resulting in the full WE-DETR model. The final model achieves 31.9% mAP and 53.2% mAP_{50}, with 43.47 M parameters and 86.59 GFLOPs. Compared with the baseline, the complete model improves mAP by +1.6% and mAP_{50} by +2.4%. Compared with the model using SPDConv but without CSPDKL, it further improves mAP by +0.5% and mAP_{50} by +0.7%. These results indicate that after shallow details are preserved and aggregated, contextual enhancement is still necessary for distinguishing small targets from complex backgrounds. The decomposed large-kernel design provides directional and long-range spatial context with moderate computational overhead, which improves the consistency of multiscale representations.

Overall, the ablation results verify the rationality of the proposed progressive compensation strategy. Plain DWT brings only limited improvement, showing that direct frequency decomposition is not sufficient. The proposed WB further improves the results, confirming the necessity of low-frequency-guided directional enhancement. The fusion module improves the interaction between structural details and semantic representations. The $\mathbf{S}_2$ frequency–spatial fusion features, SPDConv, and CSPDKL further show that compensated shallow details should be properly propagated and contextualized within the multiscale detection pipeline. Therefore, the final performance gain is not merely the result of adding more modules but comes from a coherent design that progressively preserves, enhances, fuses, and propagates fine-grained HF information for remote sensing small object detection.

3) Visualization Analysis: To provide deeper insight into how the proposed components enhance small object representations, we visualize and analyze the internal feature responses of WE-DETR from both WFC-Backbone and MSEF-Net.

As shown in Fig. 7, compared with the ResNet-backbone, WFC-Backbone produces noticeably stronger and more concentrated feature activations on small vehicles and object

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:16:19 UTC from IEEE Xplore. Restrictions apply.

contours, especially in dense traffic and parking-lot scenarios. The enhanced responses are primarily localized on target regions, while background activations are effectively suppressed. This indicates that incorporating wavelet-based frequency compensation enables the backbone to better preserve and emphasize HF structural cues that are critical for small object representation.

As shown in Fig. 8, when only conventional multiscale fusion is applied, the attention responses are relatively sparse and fragmented, and many small or weak targets receive limited activation, indicating insufficient utilization of shallow high-resolution features. After incorporating preserved $\mathbf{S}_2$ features with detail-preserving downsampling based on SPDConv, the attention maps become denser and more spatially continuous, with increased activation on small targets. But there are still some incorrect activations. With the full MSEF-Net, including multisource feature aggregation and CSP-based DLKA, the attention responses exhibit improved coverage and discrimination, enabling more effective separation between small objects and background regions.

V. Conclusion

This work proposed WE-DETR, a wavelet-enhanced detection transformer for small object detection in remote sensing imagery. By preserving HF information and strengthening multiscale feature compensation through wavelet-based decomposition, CAF, and detail-preserving downsampling, WE-DETR effectively mitigates the loss of fine-grained details during deep feature extraction. Extensive evaluations on optical, infrared, and SAR datasets demonstrate that the proposed method achieves superior small object detection performance with competitive computational efficiency, demonstrating its effectiveness on representative optical, infrared, and SAR remote sensing datasets. Despite these advances, one limitation of the current model is that it is not yet fully lightweight, and further research is needed to reduce computational complexity and parameter overhead while maintaining detection accuracy. Future work will explore model compression and efficient network design to enhance the applicability of WE-DETR in real-time or resource-constrained remote sensing scenarios.

References

[1] C. Song, Y. Zhang, M. Ma, and S. Mei, "RIFormer+: Rethinking rotation-invariant feature learning in transformer," *IEEE Trans. Multimedia*, vol. 27, pp. 8423–8435, 2025.

[2] Y. Tian, M. Zhang, J. Li, Y. Li, H. Yang, and W. Li, "FPNFormer: Rethink the method of processing the rotation-invariance and rotation-equivariance on arbitrary-oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5605610.

[3] S. Mei, C. Song, M. Ma, and F. Xu, "Hyperspectral image classification using group-aware hierarchical transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539014.

[4] J. Wang et al., "Deep high-resolution representation learning for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3349–3364, Oct. 2021.

[5] R. Zhang, "Making convolutional networks shift-invariant again," in *Proc. Int. Conf. Mach. Learn. (ICML)*, vol. 97, 2019, pp. 7324–7334.

[6] X. Ding et al., "UniRepLKNet: A universal perception large-kernel ConvNet for audio, video, point cloud, time-series and image recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 5513–5524.

[7] S. Fujieda, K. Takayama, and T. Hachisuka, "Wavelet convolutional neural networks," 2018, *arXiv:1805.08620*.

[8] T. Williams and R. Li, "Wavelet pooling for convolutional neural networks," in *Proc. Int. Conf. Learn. Represent.*, 2018.

[9] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2014, pp. 580–587.

[10] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[11] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 91–99.

[12] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. 14th Eur. Conf.* Amsterdam, The Netherlands: Springer, Oct. 2016, pp. 21–37.

[13] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[14] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis. Cham*, Switzerland: Springer, 2020, pp. 213–229.

[15] Y. Zhao et al., "DETRs beat YOLOs on real-time object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 16965–16974.

[16] Z. Zhou and Y. Zhu, "KLDet: Detecting tiny objects in remote sensing images via Kullback–Leibler divergence," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4703316.

[17] S. Ma, Y. Zhang, L. Peng, C. Sun, L. Ding, and Y. Zhu, "OWRT-DETR: A novel real-time transformer network for small object detection in open water search and rescue from UAV aerial imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4205313.

[18] X. Zhang, Y. Feng, Z. Zhao, Y. Ren, and S. Mei, "Rethinking coarse-to-fine fully convolutional transformers for salient maritime ship detection in optical remote sensing imagery," *J. Remote Sens.*, vol. 6, p. 1038, Jan. 2026.

[19] X. Zhang, Y. Feng, S. Zhang, N. Wang, G. Lu, and S. Mei, "Robust aerial person detection with lightweight distillation network for edge deployment," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5630616.

[20] H.-I. Liu, Y.-W. Tseng, K.-C. Chang, P.-J. Wang, H.-H. Shuai, and W.-H. Cheng, "A DeNoising FPN with transformer R-CNN for tiny object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4704415.

[21] X. Jiang, W. Zhang, and X. Mao, "RS-TinyNet: Stage-wise feature fusion network for detecting tiny objects in remote sensing images," 2025, *arXiv:2507.13120*.

[22] S. E. Finder, R. Amoyal, E. Treister, and O. Freifeld, "Wavelet convolutions for large receptive fields," in *Proc. Eur. Conf. Comput. Vis. Cham*, Switzerland: Springer, 2024, pp. 363–380.

[23] Y. Cao, L. Zhang, and Z. Wu, "Wavelet-based cross-modal attention network for multimedia understanding," *IEEE Trans. Multimedia*, vol. 25, pp. 3452–3466, 2023.

[24] S. He, C. Wang, and J. Li, "Multi-band wavelet enhancement network for detail-preserving visual recognition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, pp. 1842–1856, 2024.

[25] S. Yang, P. Li, and D. Xu, "Wavelet-guided multi-resolution feature fusion for medical image segmentation," *IEEE Trans. Med. Imag.*, vol. 42, pp. 3121–3134, 2023.

[26] R. Sunkara and T. Luo, "No more strided convolutions or pooling: A new CNN building block for low-resolution images and small objects," in *Proc. Joint Eur. Conf. Mach. Learn. Knowl. Discovery Databases*. Cham, Switzerland: Springer, Sep. 2022, pp. 443–459.

[27] P. Zhu et al., "Detection and tracking meet drones challenge," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 7380–7399, Nov. 2022.

[28] G. Cheng, P. Zhou, and J. Han, "Learning rotation-invariant convolutional neural networks for object detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7405–7415, Dec. 2016.

[29] J. Suo, T. Wang, X. Zhang, H. Chen, W. Zhou, and W. Shi, "HIT-UAV: A high-altitude infrared thermal dataset for unmanned aerial vehicle-based object detection," *Sci. Data*, vol. 10, no. 1, p. 227, Apr. 2023.

[30] A. Wang et al., "YOLOv10: Real-time end-to-end object detection," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 107984–108011.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:16:19 UTC from IEEE Xplore. Restrictions apply.

[31] C.-Y. Wang, I.-H. Yeh, and H.-Y. M. Liao, "YOLOv9: Learning what you want to learn using programmable gradient information," in *Proc. Eur. Conf. Comput. Vis. Cham*, Switzerland: Springer, 2024, pp. 1–21.
[32] R. Khanam and M. Hussain, "Yolov11: An overview of the key architectural enhancements," 2024, *arXiv:2410.17725*.
[33] S. Tang, S. Zhang, and Y. Fang, "HIC-YOLOv5: Improved YOLOv5 for small object detection," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2024, pp. 6614–6619.
[34] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," in *Proc. Int. Conf. Learn. Represent.*, 2020.
[35] Y. Tian, Q. Ye, and D. Doermann, "YOLOv12: Attention-centric real-time object detectors," 2025, *arXiv:2502.12524*.
[36] C. Yang, Z. Huang, and N. Wang, "QueryDet: Cascaded sparse query for accelerating high-resolution small object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 13668–13677.
[37] B. Du, Y. Huang, J. Chen, and D. Huang, "Adaptive sparse convolutional networks with global context enhancement for faster object detection on drone images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Vancouver, BC, Canada, Jun. 2023, pp. 13435–13444.
[38] F. Zhang, C. Jin, Y. Ji, J. Liu, and W. Zhang, "Evaluating RTMDet for efficient small-object detection in UAV remote sensing imagery," in *Proc. 8th Int. Conf. Comput. Inf. Sci. Appl. Technol. (CISAT)*, Jul. 2025, pp. 327–333.
[39] C. Li, R. Zhao, Z. Wang, H. Xu, and X. Zhu, "RemDet: Rethinking efficient model design for uav object detection," in *Proc. AAAI Conf. Artif. Intell.*, vol. 39, no. 5, 2025, pp. 4643–4651.
[40] S. Xu et al., "PP-YOLOE: An evolved version of YOLO," 2022, *arXiv:2203.16250*.
[41] C. Feng, Y. Zhong, Y. Gao, M. R. Scott, and W. Huang, "TOOD: Task-aligned one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3490–3499.
[42] Y. Yu, K. Zhang, X. Wang, N. Wang, and X. Gao, "An adaptive region proposal network with progressive attention propagation for tiny person detection from UAV images," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 6, pp. 4392–4406, Jun. 2024.
[43] J. Song et al., "Boundary-aware feature fusion with dual-stream attention for remote sensing small object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5600213.
[44] Y. Zhang, M. Ye, G. Zhu, Y. Liu, P. Guo, and J. Yan, "FFCA-YOLO for small object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611215.
[45] Y. Yang et al., "Small object detection in remote sensing images based on redundant feature removal and progressive regression," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5629314.
[46] Z. Wang, C. Li, H. Xu, X. Zhu, and H. Li, "Mamba YOLO: A simple baseline for object detection with state space model," in *Proc. AAAI Conf. Artif. Intell.*, vol. 39, no. 8, 2025, pp. 8205–8213.
[47] G. Zhou et al., "Lightweight multiscale feature fusion and multireceptive field feature enhancement for small object detection in the aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5640213.



Yu Song received the B.E. degree from Hainan University, Hainan, China, in 2021. He is currently pursuing the Ph.D. degree with the School of Mathematics and Statistics, Henan University, Kaifeng, China.

His research interests include intelligence information processing.



Weiguo Zuo received the B.E. degree from Henan University of Urban Construction, Pingdingshan, Henan, China, in 2024. He is currently pursuing the M.S. degree with the School of Mathematics and Statistics, Henan University, Kaifeng, China.

His research interests include intelligence information processing.



Ronghua Shang (Senior Member, IEEE) received the B.S. degree in information and computation science and the Ph.D. degree in pattern recognition and intelligent systems from Xidian University, Xi'an, China, in 2003 and 2008, respectively.

She is currently a Professor with Xidian University. Her research interests include evolutionary computation, image processing, and data mining.



Haixiao Zhuang received the B.S. and Ph.D. degrees in microelectronics and solid state electronics from the Institute of Microelectronics, Chinese Academy of Sciences, Beijing, China, in 2008 and 2010, respectively.

He is a Research Fellow at Beijing Institute of Spacecraft System Engineering, Beijing. His research interests include spacecraft system design and intelligent learning methods for space robots.



Chen Zheng (Member, IEEE) received the Ph.D. degree from the School of Mathematics and Statistics, Wuhan University, Wuhan, China, in 2012.

He is currently a Professor with the School of Mathematics and Statistics, Henan University, Kaifeng, China. He was a Post-Doctoral Researcher at the Chinese Academy of Sciences, Beijing, China, and the University of New Brunswick, Fredericton, NB, Canada. His research interests include remote sensing image segmentation and multilevel semantic analysis of remote sensing images.



Zhenzhen Liu received the B.S. degree in geographic information systems from Henan University of Urban Construction, Pingdingshan, Henan, China, in 2013, and the Ph.D. degree in remote sensing information science and technology from Henan University, Kaifeng, Henan, China, in 2023.

She is currently a Lecturer at Henan University and also a Post-Doctoral Researcher. Her research interests include remote sensing big data information extraction and processing, geospatial object and target recognition, image classification, with a particular strength in integrating artificial intelligence approaches, especially deep learning and machine learning techniques, and for the automated interpretation and information representation of complex geographic scenes.



Xiaohui Yang received the Ph.D. degree in the Institute of Intelligent Information Processing from Xidian University, Xi'an, China, in 2007.

She is currently a Professor with the School of Mathematics and Statistics, Henan University, Kaifeng, China. She was a Visiting Scholar with the University of Florida, Gainesville, FL, USA, Nanyang Technological University, Singapore, and Peking University, Beijing, China. Her research interests include intelligence information processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:16:19 UTC from IEEE Xplore. Restrictions apply.